

Assignment #01

Course: CHE 341: CHEMICAL PROCESS
CONTROL
Term: Spring 2026
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Due Date: Friday, January 23rd TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name(s) in the report. Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
 - Place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox.

 - Assignments that are not typed will not be accepted.
 - Late Assignments will not be accepted without prior approval.
 - You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report
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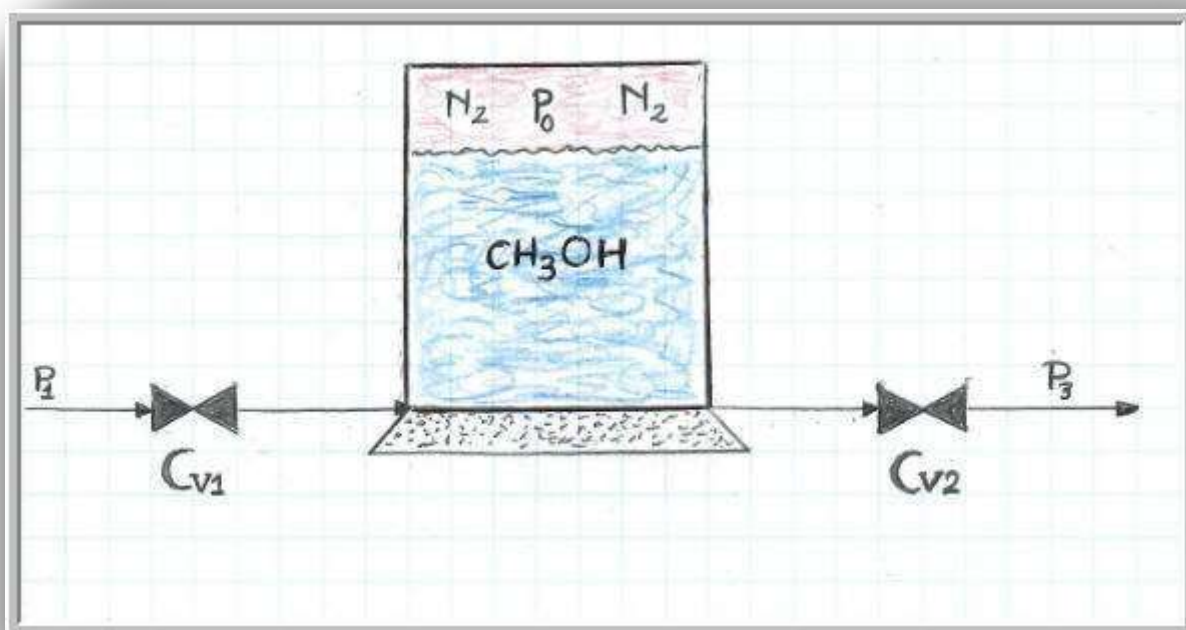
Problem A: Modeling of a Blanketed Storage Tank for Methanol

A well-insulated cylindrical tank (see Figure below) is used for storage of liquid methanol. The tank has volume of $V = 5000 \text{ ft}^3$ and the surface area of the bottom is $A = 200 \text{ ft}^2$. The liquid methanol is blanketed with nitrogen to avoid contamination with air and moisture. The feed and effluent line valve coefficients are $C_{V1} = C_{V2} = 110 \text{ gpm}/\sqrt{\text{psig}}$. The valves exhibit linear characteristics and the flow rate through the valves is given by the equations

$$F_1 = C_{V1} \sqrt{\frac{P_1}{s_g}} \quad \text{and} \quad F_2 = C_{V2} \sqrt{\frac{P_2}{s_g}}$$

where s_g is the specific gravity of methanol. The gauge pressure in the tank (measured in the nitrogen space) is $P_0 = 5 \text{ psig}$ and the temperature $T_0 = 25^\circ\text{C}$. The upstream to the feed line valve pressure is $P_1 = 33.5 \text{ psia}$ and the downstream to the effluent line valve pressure is $P_3 = 20.5 \text{ psia}$.

Figure: Methanol Storage Tank



Develop the necessary equation(s) required to monitor the liquid methanol height and the temperature of the vapor space as a function of time, due to various pressure disturbances in the feed and effluent lines. After you develop the model perform the following tasks

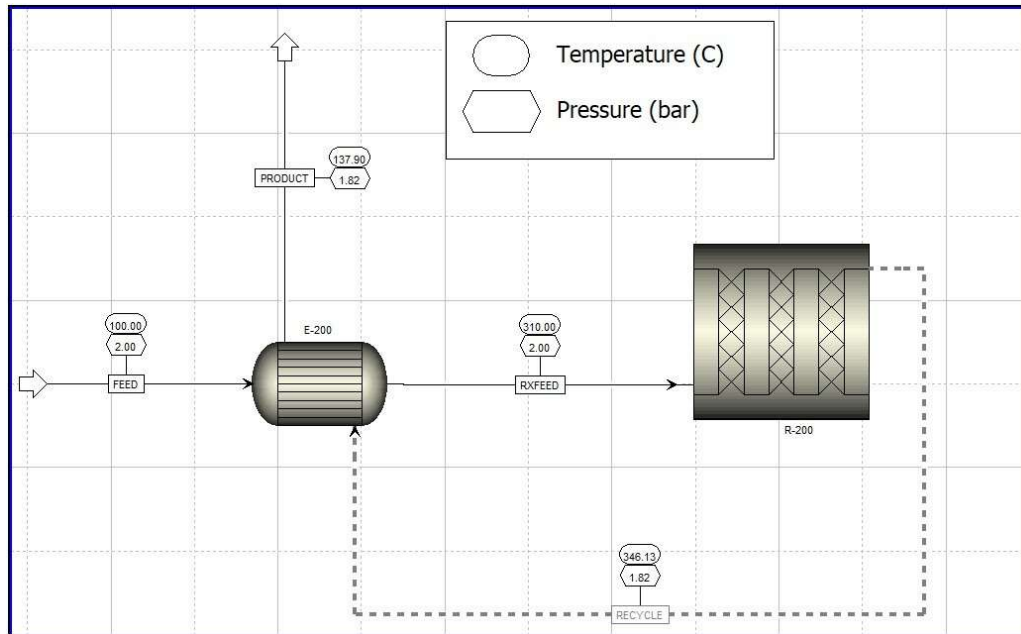
1. Calculate the steady-state liquid methanol height in ft
2. Calculate the number of moles of methanol and nitrogen in the vapor space
3. Perform a dynamic simulation for about 4 hr to demonstrate that the liquid height and temperature remain constant at the steady-state values in the absence of any disturbances
4. Start the simulation again at $t=0$ min. At time $t=2$ hr, the upstream to the feed valve pressure increases by 20%. Plot the liquid height and vapor space temperature as a function of time and calculate the new steady-state values.
5. Calculate the final pressure in **psia** of the vapor space in the storage tank
6. Calculate the time, in **min**, required to reach the new steady-state values defined as 99.5% of the asymptotic (steady-state) values.
7. In one Appendix of your report include all the plots of your analysis.
8. In a second Appendix of your report include the computer code with all the data that you used to do the numerical integration. You are free to use any software (like Matlab) but I would strongly recommend that you use the CHE Calculator

Data and Assumptions:

1. You may get the methanol and nitrogen properties from the ASPEN Plus simulator or the CHE Calculator (I would recommend the CHE Calculator)
2. Assume that nitrogen/methanol mixture in the vapor space behaves as an ideal gas with constant heat capacities C_p and C_v .
3. Assume that the amount of methanol in the vapor space does not change appreciably because of temperature and pressure variations.

Problem B: Design vs. Operational Models: Understanding Positive Feedback

Consider the ASPEN Plus simulation file that is uploaded on the UIC Blackboard along with your first assignment. The simulation file describes the catalytic oxidation of a hydrocarbon in an adiabatic fixed bed reactor. The feed to the reactor is preheated to a temperature of 310 °C using a CFE (Combined Feed Exchanger) as shown in the diagram below:



Perform the following tasks:

1. Study the simulation file and provide a description (no longer than one page) of what this design is all about with some detail on the reaction mechanism and the reactor design.
2. Modify the simulation file to reflect operational mode. The current one is the design mode.
3. Determine what will happen if due to some unforeseen disturbances the reactor effluent temperature increases by 2 °C. When answering this question assume that there are no controls in place. Describe in detail your answer and corroborate it with the proper simulations. Assuming that it takes 20 seconds for the disturbance to go through the process, plot the temperature of the reactor effluent versus time for the first 60 seconds after the onset of the disturbance.
4. Recommend a solution to remedy the situation resulting from the unexpected rise in temperature and modify the simulation file to reflect your recommendation. Describe in detail your recommendation and report the results of your simulation that reflect your recommendation. Also calculate the net increase or decrease in utilities after the onset of the disturbance.

Solution - Problem A: Modeling of a Blanketed Storage Tank for Methanol

This analysis concerns the dynamic behavior of a cylindrical methanol storage tank that is blanketed with nitrogen to prevent contamination with moisture and air. The tank exchanges liquid methanol with the surroundings through two control valves: a feed valve and an effluent valve. Changes in upstream or downstream pressures affect the liquid level within the tank, as well as the temperature and pressure of the vapor space due to gas compression and expansion. The objective of this study is to determine the steady-state operating conditions, develop a dynamic model for the coupled liquid–gas system, simulate the system response to disturbances, and quantify key dynamic performance characteristics. Throughout the analysis, methanol and nitrogen thermophysical data are obtained from the CHE Calculator, which was created by Dr Michael Carcoitas, and the nitrogen/methanol vapor mixture is assumed to behave as an ideal gas with constant heat capacities.

1. Calculate the steady-state liquid methanol height

At steady state, the liquid level in the tank is constant, so the inlet and outlet volumetric flowrates of methanol must be equal. Since both valves have the same valve coefficient and handle the same fluid, this requires the pressure drops across the feed and effluent valves to be equal. Let P_b be the pressure at the bottom of the tank. The pressure drops across the valves are

$$\Delta P_1 = P_1 - P_b, \quad \Delta P_2 = P_b - P_3$$

Setting $\Delta P_1 = \Delta P_2$ gives:

$$P_b = \frac{P_1 + P_3}{2} = \frac{33.5 + 20.5}{2} = 27.0 \text{ psia}$$

The vapor space is blanketed with nitrogen at 5 psig, which corresponds to:

$$P_{\text{vapor}} = 14.7 + 5.0 = 19.7 \text{ psia}$$

The hydrostatic pressure due to the methanol column is therefore

$$\Delta P_h = 27.0 - 19.7 = 7.3 \text{ psi}$$

Using a methanol specific gravity of $S_g = 0.806$ and the conversion $1 \text{ psi} = 2.31 \text{ ft of water}$, the steady-state liquid height is

$$h = \frac{7.3 \times 2.31}{0.806} = 20.9 \text{ ft}$$

2. Calculate the number of moles of methanol and nitrogen in the vapor space

The moles of gas in the vapor space are calculated using the ideal gas law. The total tank volume is 5000 ft^3 with a cross-sectional area of 200 ft^2 , giving a total tank height of

$$H = \frac{5000}{200} = 25 \text{ ft}$$

Using the steady-state liquid height from Question 1 ($h = 20.9$ ft), the vapor height and volume are:

$$h_v = 25 - 20.9 = 4.1 \text{ ft}$$

$$V_v = 200 \times 4.1 = 816 \text{ ft}^3$$

The vapor space is blanketed with nitrogen at 5 psig, which gives us an absolute pressure of

$$P = 14.7 + 5.0 = 19.7 \text{ psia}$$

The temperature is $25^\circ\text{C} = 537$ R. Applying the ideal gas law:

$$n_{total} = \frac{PV}{RT} = \frac{(19.7)(816)}{(10.731)(537)} = 2.79 \text{ lbmol}$$

At 25°C , the saturation vapor pressure of methanol is approximately 2.30 psia. Assuming vapor-liquid equilibrium, the partial pressures are:

$$P_{MeOH} = 2.30 \text{ psia} \quad p_{N_2} = 19.7 - 2.30 = 17.4 \text{ psia}$$

The moles of each component are:

$$n_{MeOH} = \frac{(2.30)(816)}{(10.731)(537)} = 0.33 \text{ lbmol}$$

$$n_{N_2} = \frac{(17.4)(816)}{(10.731)(537)} = 2.46 \text{ lbmol}$$

Summing the components:

$$n_{MeOH} + n_{N_2} = 0.33 + 2.46 = 2.79 \text{ lbmol}$$

The sum of the methanol and nitrogen moles match the total moles calculated from the ideal gas law.

3. Dynamic simulation with no disturbance

The purpose of this part is to verify that the tank is truly operating at steady state. To do this, the dynamic model developed earlier is simulated over time while no disturbances are applied. If the system is in a steady state, the liquid height and vapor temperature should remain constant during the simulation.

The system is described by two state variables: the liquid methanol height $h(t)$ and the vapor space temperature $T(t)$. The change in liquid height is determined from a liquid mass balance:

$$\frac{dh}{dt} = \frac{F_1 - F_2}{A}$$

Where F_1 and F_2 are the inlet and outlet flowrates through the valves. These flowrates depend on the pressure drop across the valves

$$F_1 = C_v \sqrt{\frac{P_1 - P_b}{S_g}}, \quad F_2 = C_v \sqrt{\frac{P_b - P_3}{S_g}}$$

The pressure at the bottom of the tank, P_b , is the sum of the vapor-space pressure and the hydrostatic pressure of the liquid column

$$P_b = P_g + \frac{S_g h}{2.31}$$

The vapor-space pressure P_g is calculated using the ideal gas law:

$$P_g = \frac{n_g R T}{V_v}$$

Where the vapor volume changes with liquid height according to

$$V_v = V_{tot} - Ah$$

Because the tank is well insulated and no gas enters or leaves the vapor space, the vapor temperature follows an adiabatic energy balance for an ideal gas:

$$\frac{dT}{dt} = (\gamma - 1) \frac{T}{V_v} A \frac{dh}{dt}$$

The simulation was started at the steady-state conditions found previously:

$$h(0) = 20.9 \text{ ft}, T(0) = 537 \text{ R}$$

The upstream and downstream pressures were held constant at $P_1 = 33.5$ psia and $p_3 = 20.5$ psia, and the model was integrated from $t=0$ to $t=4$ hr. The simulation shows that both the liquid height and vapor temperature remain essentially constant, with $h = 20.9$ ft and $T = 537$ R. Any small variations are due to numerical effects and do not represent physical changes, confirming steady-state operation in the absence of disturbances.

4. Dynamic simulation with a 20% increase in upstream feed pressure

To study the effect of a pressure disturbance, the dynamic simulation was restarted from the original steady-state conditions. At time $t = 2$ hr, the upstream pressure to the feed valve increased by 20%, while all other parameters and assumptions remained unchanged. The upstream pressure is held at 33.5 psia until $t = 2$ hr, after which it is increased by 20% to 40.2 psia

The system dynamics are connected to the ordinary differential equations that used previously. The liquid height is obtained from the liquid mass balance:

$$\frac{dh}{dt} = \frac{F_1 - F_2}{A}$$

Where the inlet and outlet flowrates are given by the valve equations:

$$F_1 = C_v \sqrt{\frac{P_1 - P_b}{S_g}}, \quad F_2 = C_v \sqrt{\frac{P_b - P_3}{S_g}}$$

The pressure at the bottom of the tank is:

$$P_b = P_g + \frac{S_g h}{2.31}$$

and the vapor-space pressure is calculated using the ideal gas law:

$$P_g = \frac{n_g R T}{V_v}, \quad V_v = V_{tot} - Ah$$

Because the vapor space is closed and the tank is well insulated, the vapor temperature follows an adiabatic energy balance:

$$\frac{dT}{dt} = (\gamma - 1) \frac{T}{V_v} A \frac{dh}{dt}$$

When the upstream pressure is increased at $t = 2\text{hr}$, the pressure drop across the feed valve increases, causing the inlet flowrate F_1 to exceed the outlet flowrate F_2 . This results in a rise in liquid level. As the level increases, the vapor-space volume decreases, compressing the gas and increasing both the vapor pressure and temperature until a new steady state is reached where $F_1 = F_2$

The simulation shows a smooth transient response. From the time-history plots, the new steady-state values are approximately $h_{ss,new} = 21.34\text{ ft}$ and $T_{ss,new} = 560.6^\circ\text{R}$

5. Final vapor-space pressure after the disturbance

After the 20% step increase in P_1 , the simulation gives new steady-state values of approximately $h_{ss,new} = 21.34\text{ft}$ and $T_{ss,new} = 560.6^\circ\text{R}$. Using these, the final vapor pressure is found from the ideal gas law.

$$V_{v,final} = V_{tot} - Ah_{final}$$

$$V_{v,final} = 5000 - 200 \times 21.34 = 732.8\text{ ft}^3$$

Ideal gas law for vapor pressure

The total moles in vapor space are constant from question 2: $n_g = 2.79\text{ lbmol}$ Using:

$$R = 10.731 \frac{\text{ft}^3 \text{psia}}{\text{lbmol} \times \text{R}}$$

$$P_{final} = \frac{n_g R T_{final}}{V_{v,final}}$$

$$P_{final} = \frac{2.79 \times 10.731 \times 560.6}{732.8} = 22.9\text{ psia}$$

$$P_{vapor,final} = 22.9\text{ psia}$$

6. Time required to reach 99.5% of the new steady-state values

The time required to reach the new steady state is defined as the time needed for the state variables to reach 99.5% of their total change following the disturbance at $t = 2$ hr.

For any variable $x(t)$ liquid height or temperature the 99.5% value is defined as:

$$x_{99.5} = x_0 + 0.995(x_{ss} - x_0)$$

Where x_0 is the value just before the disturbance and x_{ss} is the new steady state value

For the liquid height:

$$h_0 = 20.92 \text{ ft}, \quad h_{ss} = 21.34 \text{ ft}$$

$$h_{99.5} = 20.92 + 0.995(21.34 - 20.92) = 21.33 \text{ ft}$$

For the vapor temperature:

$$T_0 = 537 \text{ R}, \quad T_{ss} = 560.6 \text{ R}$$

$$T_{99.5} = 537 + 0.995(560.6 - 537) = 560.5 \text{ R}$$

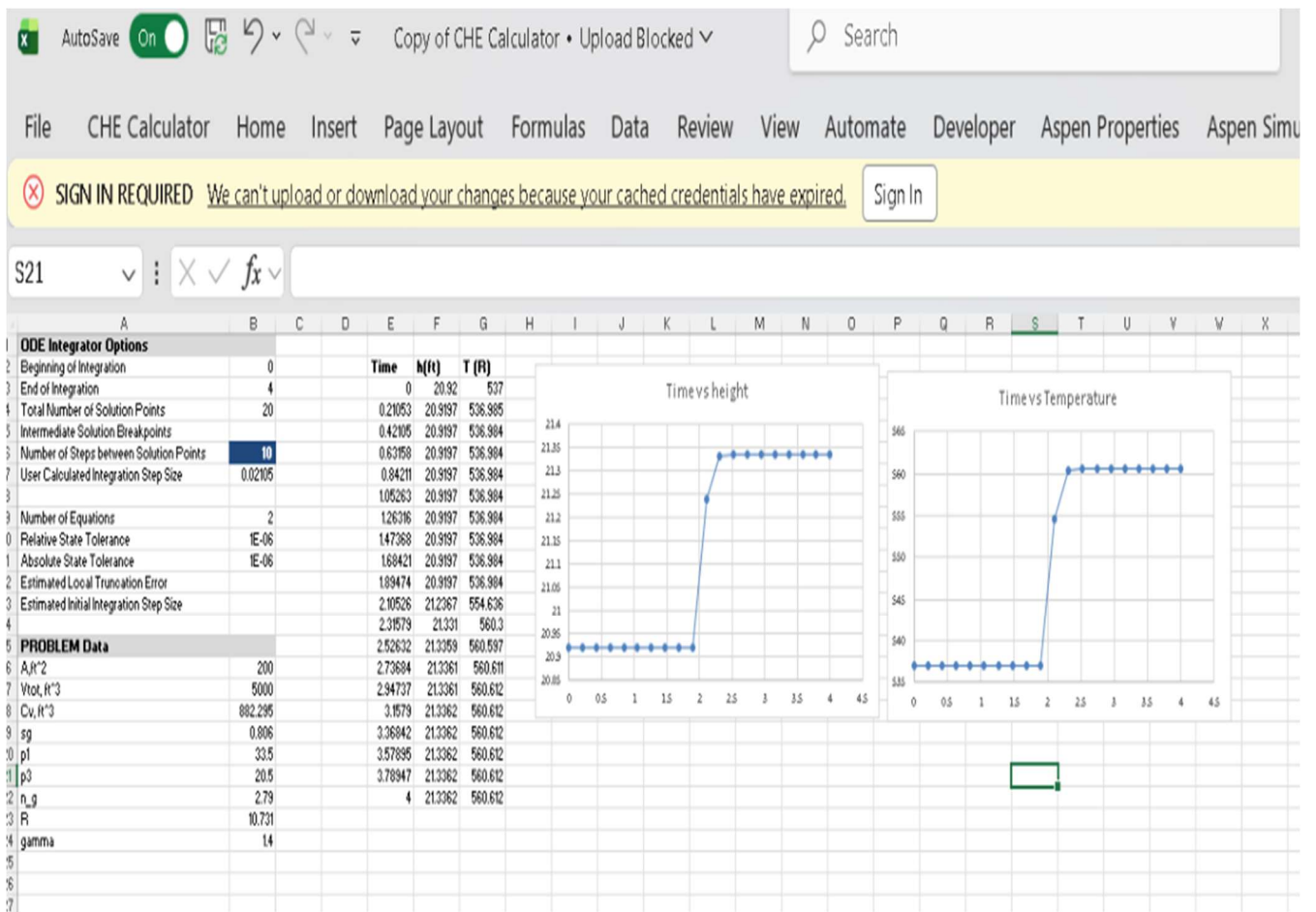
From the time-history plots, the vapor temperature reaches its 99.5% value slightly later than the liquid height and therefore controls the settling time. The temperature reaches $T_{99.5}$ at approximately $t = 2.45$ hr.

Since the disturbance occurs at $t = 2$ hr, the settling time is:

$$t_{settle} = 2.45 - 2.0 = 0.45 \text{ hr} = 27 \text{ min}$$

7. Appendix I: Plots and simulation results

AutoSave Off CHE Calculator - Read-Only - Excel										
File CHE Calculator Home Insert Page Layout Formulas Data Review View										
J9										
A B C D E F G H I J										
ODE Integrator Options										
Beginning of Integration	0			Time	h(ft)	T (R)				
End of Integration	4			0	20.92	537				
Total Number of Solution Points	20			0.210526	20.91971	536.9847				
Intermediate Solution Breakpoints				0.421053	20.91969	536.9839				
Number of Steps between Solution Points	10			0.631579	20.91969	536.9838				
User Calculated Integration Step Size	0.021053			0.842105	20.91969	536.9838				
				1.052632	20.91969	536.9838				
Number of Equations	2			1.263158	20.91969	536.9838				
Relative State Tolerance	0.000001			1.473684	20.91969	536.9838				
Absolute State Tolerance	0.000001			1.684211	20.91969	536.9838				
Estimated Local Truncation Error				1.894737	20.91969	536.9838				
Estimated Initial Integration Step Size				2.105263	21.23666	554.6362				
				2.315789	21.33104	560.3003				
PROBLEM Data				2.526316	21.33589	560.5966				
A, ft ²	200			2.736842	21.33613	560.6114				
Vtot, ft ³	5000			2.947368	21.33614	560.6121				
Cv, ft ³	882.295			3.157895	21.33615	560.6121				
sg	0.806			3.368421	21.33615	560.6121				
p1	33.5			3.578947	21.33615	560.6121				
p3	20.5			3.789474	21.33615	560.6121				
n_g	2.79			4	21.33615	560.6121				
R	10.731									
gamma	1.4									



Height and temperature vs time results

8. Appendix II: VBA code

Appendix 2 has the VBA Codes for FODEA1 and FODEA2

(General) ▾	TankODE
Option Explicit	
<pre> Private Function ClampNonNeg(ByVal x As Double) As Double If x < 0# Then ClampNonNeg = 0# Else ClampNonNeg = x End If End Function </pre>	
<pre> Function TankODE(ByVal t As Double, ByRef x() As Double, Optional ByVal prn As Range) As Double() ' x(1) = liquid level h (ft) ' x(2) = vapor temperature Tv (°R) Dim hLvl As Double, tVap As Double hLvl = x(1) tVap = x(2) ' ---- Parameters (read once) Dim areaFt2 As Double, volTot As Double, cvEq As Double, spGr As Double Dim pUp As Double, pDown As Double, nMol As Double, R As Double, k As Double With prn areaFt2 = .Cells(1, 1).Value2 volTot = .Cells(2, 1).Value2 cvEq = .Cells(3, 1).Value2 spGr = .Cells(4, 1).Value2 pUp = .Cells(5, 1).Value2 pDown = .Cells(6, 1).Value2 nMol = .Cells(7, 1).Value2 R = .Cells(8, 1).Value2 k = .Cells(9, 1).Value2 End With ' ---- Geometry / pressures Dim vapVol As Double vapVol = volTot - areaFt2 * hLvl If vapVol <= 0.0000000000001 Then vapVol = 0.000000000001 ' avoid divide-by-zero / negative volume Dim pGas As Double pGas = (nMol * R * tVap) / vapVol ' psia, ideal gas Dim pBot As Double pBot = pGas + (spGr * hLvl / 2.31) ' add hydrostatic head (psi) ' ---- Valve ?P (no reverse flow) Dim dpIn As Double, dpOut As Double dpIn = ClampNonNeg(pUp - pBot) dpOut = ClampNonNeg(pBot - pDown) ' ---- Flows (ft^3/hr) Dim qIn As Double, qOut As Double qIn = cvEq * Sqr(dpIn / spGr) qOut = cvEq * Sqr(dpOut / spGr) ' ---- ODEs Dim dhdt As Double, dVdt As Double, dTdt As Double dhdt = (qIn - qOut) / areaFt2 dVdt = -areaFt2 * dhdt ' Temperature model: ' A) Isothermal (default): dTdt = 0# ' B) If you want isentropic ideal gas instead, comment the line above and use: ' dTdt = -(k - 1#) * tVap * (dVdt / vapVol) ' ---- Return derivative vector Dim dx() As Double ReDim dx(1 To 2) dx(1) = dhdt dx(2) = dTdt TankODE = dx End Function </pre>	

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Function fODEA2(T As Double, u() As Double, Optional Param As Range) As Double()
    Dim h As Double, Tv As Double
    h = u(1)          ' liquid height, ft
    Tv = u(2)         ' vapor temperature, °R

    ' Parameters from worksheet
    Dim A As Double, Vtot As Double, CvStar As Double, sg As Double
    Dim P1 As Double, P3 As Double, nGas As Double, Rgas As Double, gamma As Double

    A = Param.Cells(1, 1)
    Vtot = Param.Cells(2, 1)
    CvStar = Param.Cells(3, 1)
    sg = Param.Cells(4, 1)

    ' --- time-dependent disturbance in P1 ---
    If T < 2# Then
        P1 = Param.Cells(5, 1)          ' old P1 = 33.5 psia
    Else
        P1 = 1.2 * Param.Cells(5, 1)    ' +20% step to 40.2 psia
    End If
    ' -----

    P3 = Param.Cells(6, 1)
    nGas = Param.Cells(7, 1)
    Rgas = Param.Cells(8, 1)
    gamma = Param.Cells(9, 1)

    ' Local variables
    Dim Vv As Double, Pgas As Double, Pbottom As Double
    Dim dP1 As Double, dP2 As Double
    Dim F1 As Double, F2 As Double
    Dim dhdt As Double, dVvdt As Double, dTdt As Double
    Dim f() As Double
    Dim n As Integer

    ' Vapor volume
    Vv = Vtot - A * h

    ' Gas pressure
    Pgas = nGas * Rgas * Tv / Vv

    ' Bottom pressure (include hydrostatic head)
    Pbottom = Pgas + sg * h / 2.31

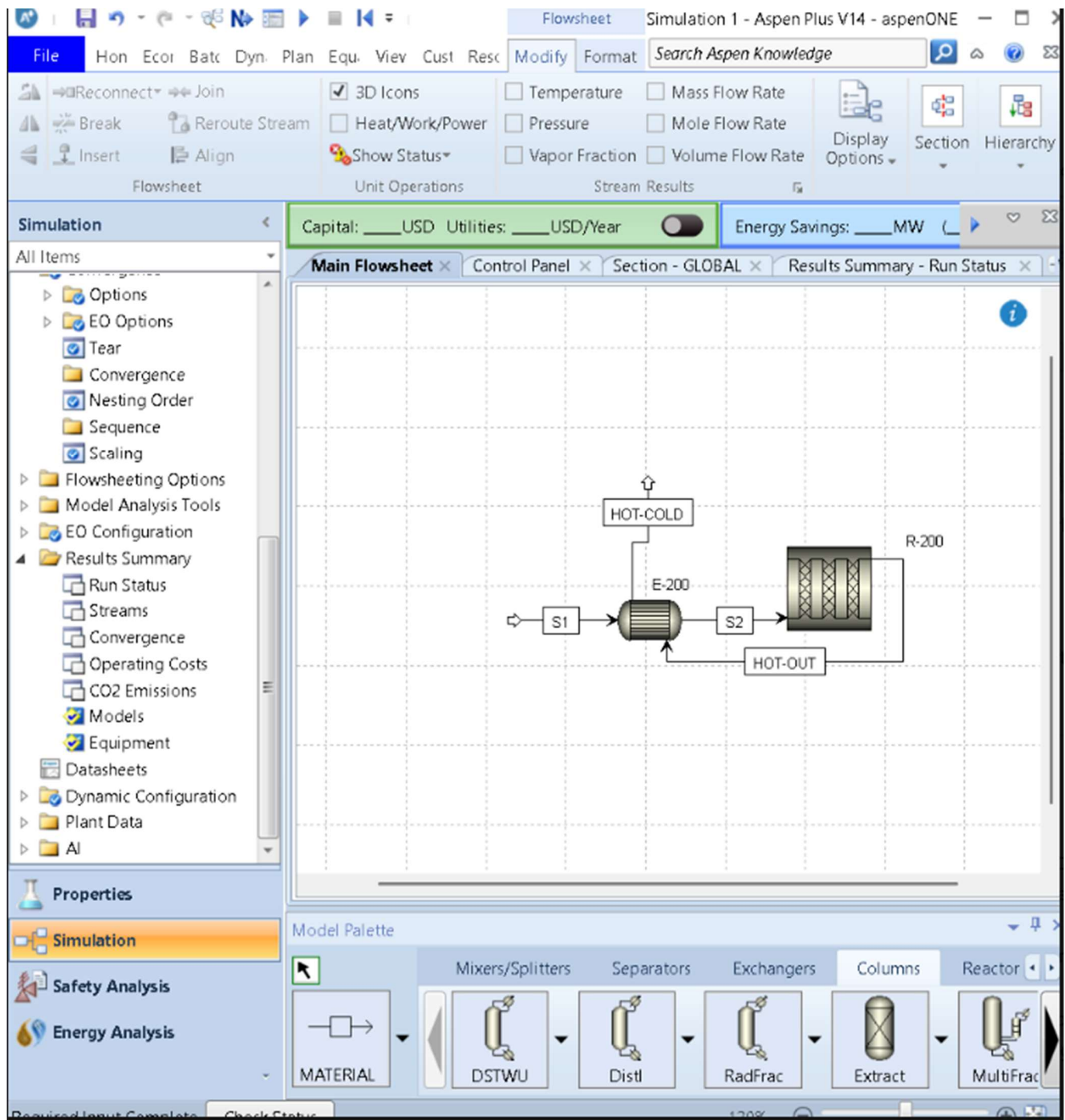
    ' Valve pressure drops
    dP1 = P1 - Pbottom
    dP2 = Pbottom - P3
    If dP1 < 0 Then dP1 = 0
    If dP2 < 0 Then dP2 = 0

    ' Valve flows (ft³/hr)
    F1 = CvStar * Sqr(dP1 / sg)
    F2 = CvStar * Sqr(dP2 / sg)

    ' ODEs
    dhdt = (F1 - F2) / A
    dVvdt = -A * dhdt
    dTdt = -(gamma - 1#) * (Tv / Vv) * dVvdt

    ' Return derivative vector
    n = UBound(u)
    ReDim f(1 To n) As Double
    f(1) = dhdt
    f(2) = dTdt
    fODEA2 = f
End Function

```



The Aspen Plus simulation did not initially converge because the process includes an adiabatic exothermic reactor coupled with a feed effluent heat exchanger, which creates a strong recycling loop. Small changes in reactor outlet temperature affected the feed preheat, causing numerical instability and preventing convergence. The problem was made worse when internal recycling

streams were over-specified and when the reaction stoichiometry was not fully balanced. To fix this, all internal and outlet streams were left unspecified and only the feed stream was defined. The reaction was corrected to ensure proper mass balance, and a tear stream was added on the hot-side outlet of the heat exchanger to help the solver iterate the recycle loop. Even with a defined tear stream and corrected reaction stoichiometry, the solver was unable to reach a steady solution under the specified design conditions, indicating that the process configuration is inherently difficult to operate without additional temperature control.